

SEMINAR REPORT

on

**Wind Profile Estimation for K-Band Doppler Radar Based
on Mutual Convolution Cost Function Method**

Submitted by

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CERTIFICATE

*Certified that the seminar report entitled “Wind Profile Estimation for K-Band Doppler Radar Based on Mutual Convolution Cost Function Method” is a bonafide work of **Roopesh O R (20323085)** towards the partial fulfillment for the award of the degree of B.Tech in Electronics and Communication of Cochin University of Science and Technology, Kochi-682022.*

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ABSTRACT

K-band Doppler wind profile radars are widely used to estimate atmospheric wind velocity and direction; however, weak atmospheric echoes are often corrupted by noise, clutter, and multiple Doppler spectral peaks under low signal-to-noise ratio (SNR) conditions. This paper proposes a Mutual Convolution Cost Function (MCCF) algorithm to improve the robustness of wind-profile estimation. The method preprocesses the radar echoes, adaptively constructs a space-time Doppler window based on wind continuity and wind-shear constraints, applies Mutual Convolution with multiple delay times to suppress uncorrelated noise while preserving coherent atmospheric echoes, and employs an improved Mutual Convolution Complex Least Mean Square (MCLMS) adaptive filter for further clutter suppression. Candidate Doppler peaks are then evaluated using a cost function that jointly considers spectral power and velocity continuity to identify the most probable atmospheric echo. Experimental evaluation using real VORTRAD K-band radar data demonstrates that the proposed method improves echo SNR by approximately 2–4 dB, significantly reduces false spectral peaks, produces smoother and more physically consistent wind profiles, and achieves correlation coefficients of approximately 0.95 with collocated LiDAR measurements, outperforming the AME, MPCF, and IME methods under low-SNR and multiple-peak conditions.

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List of Abbreviations

AME:	Adaptive Moment Estimation
CA:	Coherent Accumulation
CLMS:	Complex Least Mean Square
DC:	Direct Current
FFT:	Fast Fourier Transform
IME:	Improved Moment Estimation
MCCF:	Multiple Convolution Cost Function
MCLMS:	Mutual Convolution and Complex Least Mean Square
MPCF:	Multiparameter Cost Function
PRI:	Pulse Repetition Interval
SNR:	Signal-to-Noise Ratio
ST-W:	Space-Time Doppler Window

CHAPTER 1

INTRODUCTION

The research paper titled “*Wind Profile Estimation for K-Band Doppler Radar Based on Mutual Convolution Cost Function Method*” proposes a new method to estimate wind profiles by combining several methods including MCCF. The paper primarily focuses K-Band radar and on its potential application of such as efficient wind energy generation as well as challenges posed by seasonal atmospheric variation in measurement of near surface (< 1 km) wind turbulence.

1.1 Problem Addressed

The primary problem is that radar echoes from atmospheric turbulence are often heavily corrupted by strong background noise, resulting in a low SNR. This issue worsens in colder environments where decreased air density further weakens the returning signal. Additionally, the circular detection area of the radar antenna (widening of beam in higher altitudes) causes the returning echoes to frequently contain multiple spectral peaks. This makes it difficult to isolate the true frequency offset associated with the actual wind target.

Traditional data processing methods attempt to extract the wind profile by analyzing coherent accumulation (CA) signals with a cost function. While these methods work well when the SNR is high, their accuracy deteriorates under low SNR conditions with multiple competing peaks. Under these constraints, traditional algorithms become highly susceptible to clutter and noise, often mistaking false peaks for the true wind signal, which degrades the overall accuracy of the radar.

1.2 Overview of Process

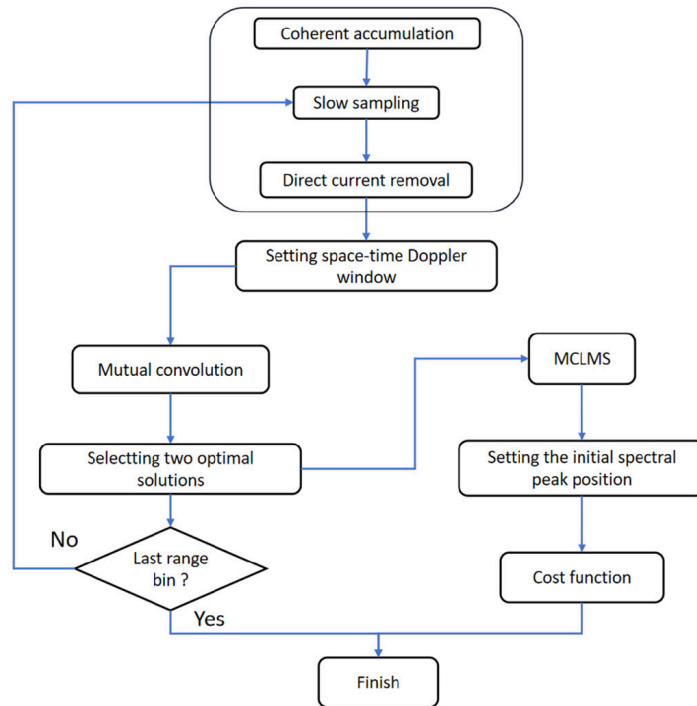
The paper describes a 5 step process to extract wind velocity:

1. **Signal Preprocessing:** Performs CA which boosts the steady wind signal and averages out random background noise. Resamples data into specific altitude slices (range bins) to track temporal changes in wind speeds at different heights. Finally atmospheric clutter and ground reflections are removed.
2. **Space-Time Doppler Window:** The system uses recent measurements to set a window in both spacial and temporal range. Any frequency peaks that fall far outside this window are ignored.

3. **Mutual Convolution:** The system takes an initial segment of radar signal at a height and compares it against slightly delayed segments of the same signal. It applies convolution with them and because random noise does not repeat exactly over time, it is canceled out. The stable wind signal persists across the segments and is amplified.
4. **Modified Complex Least Mean Square (MCLMS) Filter:** It minimizes the difference between the desired signal and the actual output, further smoothing the data and eliminating leftover clutter.
5. **Cost Function Evaluation:** The final step chooses the correct wind speed from any remaining signal peaks, by assigning penalty score to different aspects of the peaks and selecting the one with least penalty.

Finally the results are validated using data from a LIDAR operating in the same time period. The overall process is summarized in Figure 1.1

Figure 1.1: Overall flow of the method



1.3 Experiment Setup

The authors have utilized the K-band vortex radar (VORTRAD) as experiment equipment provided by Suzhou Dufeng Technology Company, which estimates wind profile by detecting Doppler frequency offset generated by atmospheric turbulence. The specification

of VORTRAD system is given in Table 1.1. All experiments are conducted at Changxing Island, China.

The experiment setup is shown in Figure 1.2. The VORTRAD is equipped with a shield and two devices are placed 3 m apart. The distance resolution, velocity resolution, and measurement distance of the two devices are consistent, and the accuracy of this LIDAR under clear sky conditions reaches 99%. For comparing the effectiveness of proposed method, researchers have compared MCCF against CA, AME, MPCF, and IME using SNR and Correlation coefficient

Table 1.1: Specification of VORTRAD Radar

Number	Specification	Value
1	Frequency band	K-band
2	Operation mode	Single frequency pulse
3	Number of beams	4 (East, South, West, North)
4	Tilt Angle	15°
5	Pulse period	2us
6	Pulse width	≈ 133.3ns
7	No.of coherent accumulation	50
8	Slow sampling points	2048
9	FFT points	1024
10	Maximum range	300m
11	Window coefficient	2
12	Maximum radical velocity	30 m/s
13	Radial velocity resolution	0.06 m/s
14	Range resolution	20m

Figure 1.2: Experiment setup



1.4 Earlier Works

Earlier research on wind-profile radar signal processing has primarily focused on reliable Doppler spectral peak detection, wind-profile tracking, and suppression of noise and clutter. Clothiaux et al. [1] proposed identifying atmospheric spectral peaks by constructing possible peak chains and selecting the longest chain with the maximum sum of spectral power, exploiting the expected continuity of wind velocity with height. Anandan et al. [2] introduced Adaptive Moment Estimation (AME), which uses SNR and wind shear information to improve the identification of atmospheric echoes. Sinha et al. [3], [4] developed the Multi-parameter Cost Function (MPCF) method, which evaluates candidate Doppler peaks using multiple parameters such as spectral power and velocity/wind-shear continuity rather than simply selecting the strongest peak. Li et al. [5] proposed an Improved Moment Estimation (IME) method to enhance wind estimation, while Bhatta et al. [6] introduced Viterbi Data Association (VDA) to track the most probable wind-velocity profile across successive range bins. These methods use the spatial continuity of atmospheric winds, but their performance can become limited when the radar echo has low SNR and multiple competing spectral peaks.

For turbulence and weak-echo detection, Coherent Accumulation (CA) is traditionally used to improve SNR by coherently integrating multiple radar pulses. Several studies have subsequently investigated adaptive filtering techniques for further noise and clutter suppression. Took and Mandic [7] proposed Quaternion Least Mean Square (QLMS) based on CLMS for wind-vector estimation. Pei and Ding [8] introduced the Fractional Fourier Transform (FRFT) to concentrate the energy of unstable signals, thereby improving adaptive-filtering performance. Chen et al. [9] combined FRFT-domain processing with Normalized Leakage LMS (NL-LMS) to enhance target-signal energy and suppress clutter. Shi et al. [10] proposed Shrinkage Widely Linear CLMS (SWL-CLMS) to improve convergence speed and reduce steady-state misalignment. Menguc and Acir [11] developed an Augmented Complex-Valued Least Mean Kurtosis (ACLMK) filter that uses augmented statistics and the kurtosis of the complex error signal as its cost function. Zhang et al. [12] proposed a Widely Linear Complex Estimation Input Adaptive Filter (WLC-EIAF) for unbiased signal estimation when both input and output are affected by noise. Building on these developments, the paper combines the advantages of existing moment-estimation, cost-function, and adaptive-filtering approaches with Mutual Convolution to form the proposed MCCF framework, aiming specifically at reliable wind-profile extraction in low-SNR conditions with multiple Doppler spectral peaks.

CHAPTER 2

RESEARCH METHODOLOGY

This section introduces the data processing algorithm in detail and mainly addresses the issues of low SNR and multiple spectral peaks by the MCCF algorithm. The algorithm is divided into five parts, signal preprocessing, setting the space-time Doppler window, mutual convolution, MCLMS filter, and generating the cost function.

2.1 Signal Preprocessing

This phase prepares data for MCCF algorithm. In this phase, 3 operations are performed on data:

2.1.1 Coherent Accumulation (CA)

Here atmospheric turbulence is characterized as a weak target. The model assumes that turbulence moves at a uniform velocity over a short period. Hence its echo is akin to a stable periodic signal. Conversely, noise exhibits instability and lacks periodicity. The CA adds and averages short segments of radar signals thereby reducing noise strength and amplifying turbulence echo. Throughout the paper 2-D matrix generated by the result after N-order CA is represented as:

$$X = [x_{c1}^T, x_{c2}^T, \dots, x_{c(N-1)}^T, x_{cN}^T]$$

Where x_{ci} is result of i^{th} CA. (Hence the column vector of the X represents the result of one accumulation)

2.1.2 Slow Sampling

To detect the motion state of turbulence at different altitudes, we need to extract data from a specific range bin. When a pulse is transmitted, the radar receives echoes from all range bins (heights). After one Pulse Repetition Interval (PRI), it transmits the next pulse and repeats the process. For a single range gate (height), the radar stores one complex echo sample from every transmitted pulse. This sequence is called *slow-time signal*. To obtain this, a row vector from X is used. This process is called *slow sampling* in the distance dimension. The slow sampling frequency must satisfy the Nyquist criterion in order to avoid blurring the maximum turbulence velocity, and the slow sampling frequency $f_{d\min}$ can be given as

$$f_{d\min} = \frac{4v_{\max}}{\lambda}$$

where $v_{\max}$ represents the maximum velocity range and λ represents the work wavelength of the radar

2.1.3 DC and Clutter Removal

Hardware components and ground reflections create direct current (DC) interference, which shows up as clutter near the zero-frequency point. In case of VORTRAD which adopts a zero-intermediate-frequency RF transceiver architecture, produces signal containing a large DC power, and ground clutter with a small frequency offset. The paper doesn't mention a particular clutter removal method.

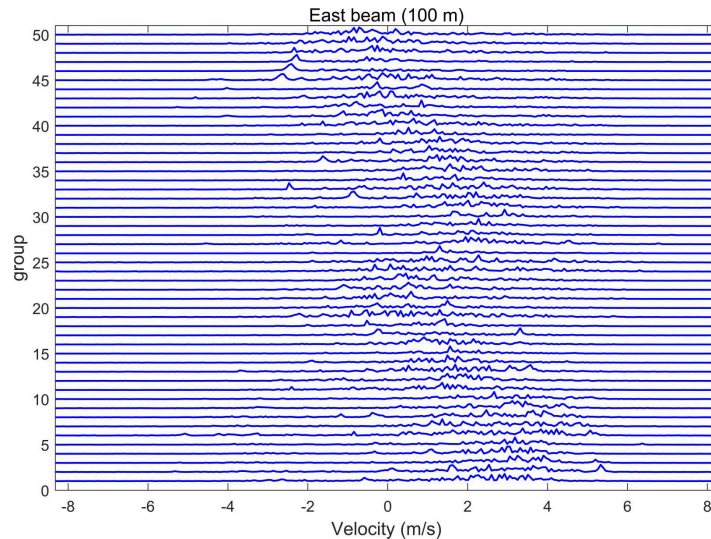
2.2 Setting the Space-Time Doppler Window

In this stage, the search window for peaks is reduced by using 2 property of wind velocity:

- Turbulent motion state between adjacent altitudes continuously changes and there is a maximum wind shear associated with a difference in altitude (typically less than 0.5 m/s in the range of 10 m and is less than 1 m/s in the range of 20 m).
- Wind velocity exhibits continuity over multiple pulse same altitude as shown in Figure 2.1

Therefore, when setting this search window (doppler window), it is essential to consider both spatial and temporal aspects. The size of doppler window.

Figure 2.1: Wind velocity over 50 consecutive groups at 100 m. (provided by researchers)



The researchers have quantitatively modeled the window size using frequency offset corresponding to wind shear (d_f) and “window coefficient related to altitude interval” (α). First,

considering space continuity, assuming that the spectrum estimation result of the previous range bin is F_0 , the space Doppler window can be represented as

$$W_1 = (F_0 - \alpha d_f, F_0 + \alpha d_f)$$

Where d_f is given by:

$$d_f = \frac{V_d}{d_v} = \frac{V_d}{f_{dmin} \lambda / L}$$

Where V_d = maximum value of wind shear and L = number of points in FFT.

Now, considering time continuity, and assuming the previous spectrum estimation result at the same range bin is T_0 , the time Doppler window can be represented as

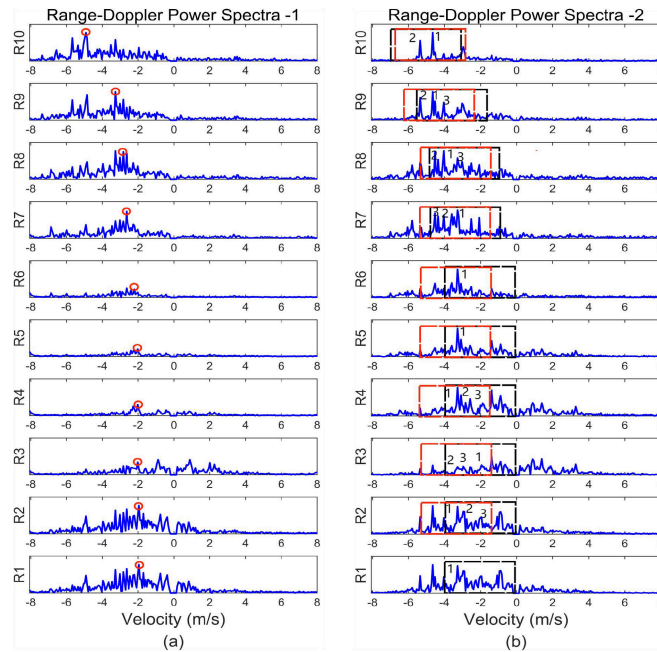
$$W_2 = (T_0 - \alpha d_f, T_0 + \alpha d_f)$$

The window used for peak estimation is intersection of these 2 windows, the researchers have termed this window as *space-time Doppler window*, expressed as:

$$ST - W = \begin{cases} W_1 \cap W_2 & \text{if } W_1 \cap W_2 \geq \alpha d_f \\ W_1 & \text{else} \end{cases}$$

An example case of ST-W and peak estimation is illustrated in Figure 2.2

Figure 2.2: (a) spectra peak selection highlighted in red circles (b) Space (red) and Time (black) doppler windows marked



2.3 Mutual Convolution

To minimize the error rate of spectral estimation, efforts should be made to attenuate noise and retain significant spectral peaks before identifying candidate spectral peaks. The main idea is to exploit the temporal correlation of atmospheric echoes while suppressing random noise which is highly uncorrelated. Suppose successive pulses observe the same moving air parcel. The received signal changes slowly. Similarly, its phase also changes smoothly. Therefore, successive slow-time samples are strongly correlated.

Because of this difference, the researchers used mutual convolution to measure the similarity between delayed versions of the slow-time signal.

The method is described below:

First, take a row vector $P = (p_1, p_2, \dots, p_N)$ of one range bin from the preprocessing data X . Then, take out vectors with equal length from P to form a new row vector. Here the researchers used half the size of the original vector as new vector. Let these constructed vectors be:

$$V_{r1} = (p_1, p_2, \dots, p_{N/2})$$

The half length for sequences allows several overlapping signal segments to be created. The paper defines number of delay points I as:

$$I = \frac{N}{2(n-1)}$$

where

- N = total length of vector P
- n = desired number of extracted vectors

From this a new matrix can be given as:

$$V = [V_{r1}, V_{r1}, \dots, V_{rn}]$$

Now mutual convolution is performed:

$$\begin{aligned} V_{C1} &= V_{r2} * V_{r1}^{\text{HR}} \\ V_{C2} &= V_{r3} * V_{r1}^{\text{HR}} \\ &\vdots \\ V_{C(n-1)} &= V_{rn} * V_{r1}^{\text{HR}} \end{aligned}$$

Where HR means that the conjugate and then reverse order. After the above calculation, the number of convolution results is $n - 1$ and the length of each calculation result becomes $N - 1$. To maintain the same time complexity and velocity resolution of calculation, a new matrix can be generated by extracting data from the middle of the vector, with the number being $N/2$, given as:

$$[V'_C] = \begin{bmatrix} V'_{C1} \\ V'_{C2} \\ \vdots \\ V'_{C(n-1)} \end{bmatrix} = \begin{bmatrix} V_{C1} \left(\frac{N}{4} : \frac{3N}{4} \right) \\ V_{C2} \left(\frac{N}{4} : \frac{3N}{4} \right) \\ \vdots \\ V_{C(n-1)} \left(\frac{N}{4} : \frac{3N}{4} \right) \end{bmatrix}$$

Then maximum of power spectra of each row is computed:

$$V'_{cf} = [\max\{|\mathcal{F}([V'_c])|\}]$$

where $\mathcal{F}$ represents Fourier transform. The maximum of V'_{cf} corresponds to some optimal solution X_r (let that be V'_{Ck}). In practice the researchers choose to keep adjacent readings ($V'_{C(k-1)}$ and $V'_{C(k+1)}$) which are kept as candidate peaks or suboptimal solutions X'_r .

2.4 MCLMS

MCLMS is a adaptive filter. Its purpose is to further suppress residual clutter and random noise after the Mutual Convolution stage has identified the most coherent signal segment. The algorithm uses the optimal convolution result X_r and the adjacent suboptimal convolution result X'_r as the input and desired signals of a CLMS adaptive filter. The underlying assumption is that these two signals contain the same atmospheric echo because they originate from adjacent delay windows, while their noise and clutter components are largely uncorrelated. By adapting the filter to transform X_r into X'_r , the filter reinforces the common atmospheric component and suppresses the components that differ between the two signals. Unlike a conventional adaptive filter, where adaptation and filtering occur simultaneously, the proposed MCLMS algorithm separates these operations into two stages. During the first stage, the filter is used only for training – the adaptive filter coefficients are continuously updated until they converge. After convergence, the learned coefficients are copied and reused in a second filtering pass to generate the final filtered output. The specific calculation

process is as follows. The optimal solution is represented as an $\frac{N}{2} - N/2$ - dimensional row vector and the length of the coefficient filter is denoted as k . First a normalization factor is calculated.

$$u = \frac{1}{\sum_{j=1}^{N/2} |X_r(j)|^2}$$

This balances how coefficients are adjusted even when one input is much larger in comparison to others. The adaptive filter has a length of k coefficients. For every sample position j , an input vector X is constructed from the samples of X_r . If there are insufficient previous samples at the beginning of the sequence, zeros are inserted.

$$X = \begin{cases} \text{zeros}(1, k - j), X_r(1 : j) & , j < k \\ X_r(j - k + 1 : j) & , j \geq k \end{cases} \quad j \in \{1, 2, \dots, N/2\}$$

Where k = length of “filter coefficient vector”. This effectively forms a sliding window over the signal, allowing the adaptive filter to operate as a finite impulse response (FIR) filter. The filter coefficients are initially set to zero: $W_1 = 0$

For each input vector, the filter output is calculated using

$$Y_j = W_1 X^T$$

which is simply the dot product between the current coefficient vector and the input vector. The difference between the desired signal X'_r and the filter output produces the estimation error,

$$e_j = X'_{rj} - Y_j$$

The coefficients are then updated using

$$W'_1 = 0.98W_1 + ue_j X^H$$

The factor 0.98 is a leakage factor that slightly reduces the magnitude of the coefficients during each iteration, preventing them from growing excessively and improving numerical stability

The coefficient update process is repeated for every sample in the signal until all data have been processed. At this stage, the coefficients have converged to values that best represent the common characteristics shared by X_r and X'_r . Here we let

$$W_2 = W'_1$$

In the second pass, the algorithm repeats the filtering operation using the fixed coefficient vector W_2 but without further adaptation. Since the coefficients have already converged, the filter produces a stable output Y_r , which contains a stronger atmospheric echo and significantly reduced noise and clutter. This filtered signal is subsequently used by the MCCF stage for reliable Doppler peak identification and wind profile estimation.

2.5 Generating the Cost Function

After the Mutual Convolution and MCLMS stages, the radar signal has been significantly cleaned, and a Doppler spectrum has been obtained for every range bin. However multiple spectral peaks may still exist within a single Doppler spectrum. In conventional spectral estimation methods, it is generally assumed that the first range bin contains only one dominant spectral peak, which is selected as the atmospheric echo. The researchers point out that this assumption is not valid for the VORTRAD and other wind profilers because strong ground clutter, interference, and other unwanted echoes often produce multiple prominent peaks in the first range bin. Consequently, selecting the strongest peak directly can result in incorrect wind estimation. To overcome this problem, the paper introduces the Mutual Convolution Cost Function (MCCF), which evaluates candidate spectral peaks using both their spectral power and their physical consistency with neighbouring range bins.

first step is to calculate the average value corresponding to the power spectrum of the first range bin, it can be given as:

$$M_1 = \frac{1}{n-1} \sum_{i=1}^{n-1} \mathcal{F}(V'_{Ci})$$

The resulting average spectrum provides a more stable estimate of the true Doppler spectrum in the first range bin. The algorithm then identifies the strongest peak in this averaged spectrum,

$$M_{1,1} = \max |M_1|$$

which serves as the initial reference for subsequent peak tracking through higher range bins. For every range bin, the Doppler spectrum is then computed from the MCLMS filtered signal, $M_r = |\mathcal{F}(Y_r)|$

To reduce the possibility of selecting insignificant noise peaks, the algorithm first sorts all spectral peaks according to their power and retains at most three candidate peaks within each ST-W. Furthermore, a candidate peak is considered only if its power satisfies

$$M_{r,j} \geq 0.4M_{r\max}$$

Where $M_{r\max}$ is maximum spectral peak within the ST-W and $M_{r,j}$ represents the power value corresponding to the j th spectral peak of the r th range bin. This threshold removes weak noise peaks. The algorithm then evaluates how well each candidate peak agrees with the wind velocity estimated in the previous range bin. Atmospheric winds normally vary smoothly with altitude, and therefore the Doppler velocity should also change gradually between adjacent range bins. To quantify this behaviour, the paper defines a velocity cost

$$V_{r+1,j} = 1 - \left| \frac{(M_{r+1} - M_r)d_v}{\alpha V_d} \right|$$

where

- d_v = velocity resolution
- V_d = maximum wind shear
- α = window coefficient,
- $V_{r+1,j}$ = velocity cost value corresponding to the j th

spectral peak of the $(r + 1)$ th range bin.

Finally, the algorithm combines two independent criteria to evaluate each candidate peak – velocity continuity (represented by velocity cost $V_{r+1,j}$) and spectral power of the candidate peak ($M_{r+1,j}$). These two constraints are combined through the MCCF cost function:

$$F_{r+1,j} = w_1 V_{r+1,j} + w_2 M_{r+1,j} \quad \text{and } w_1 + w_2 = 1$$

where w_1 and w_2 are weighting factors controlling the relative importance of velocity continuity and spectral power. This weighted cost function ($F_{r+1,j}$) forces selection of candidate peak that simultaneously possess sufficient spectral power and reasonable continuation of the wind profile. The peak with the highest cost value is selected as the atmospheric Doppler peak for that range bin, and the process is repeated for all subsequent range bins until the complete wind profile has been estimated.

CHAPTER 3

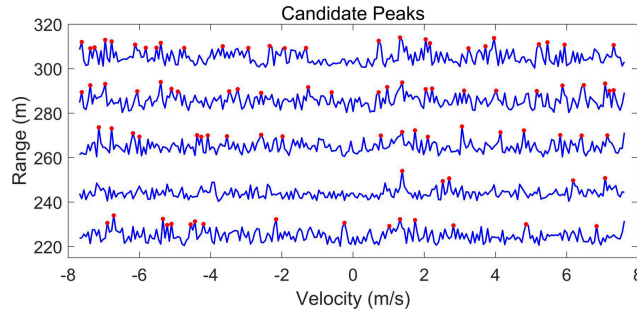
RESULTS

The authors have evaluated the algorithm in four ways: Delay time for mutual convolution, effectiveness in reducing clutter and improving SNR, comparing wind profile estimate with existing algorithm, validating the estimated wind velocities against LiDAR using correlation coefficient. Each experiment methods gives a different aspect of the proposed algorithm.

3.1 Performance Evaluation at Different Delay Times

This experiment investigates how the delay time used in the Mutual Convolution stage affects clutter suppression. The algorithm correlates the original signal with delayed versions of itself. If the delay is too short, the clutter and noise remain highly correlated and are not sufficiently suppressed. If the delay is too large, even the atmospheric echo begins to decorrelate, reducing the effectiveness of the filtering.

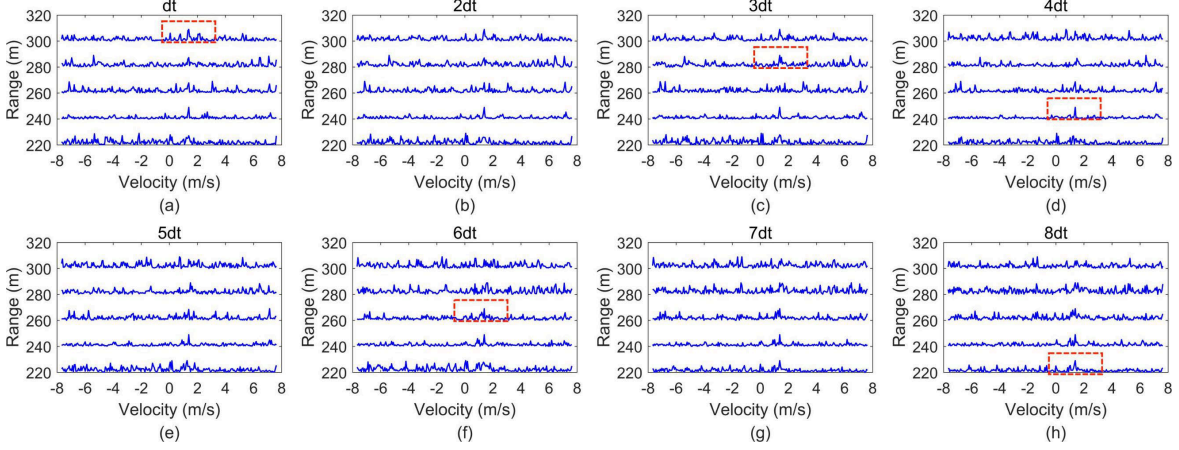
Figure 3.1: Range-Doppler power spectra after preprocessing



The researchers begin by showing the preprocessed Doppler spectra in Figure 3.1. These spectra correspond to echoes between 220 m and 300 m altitude. The red dots represent all spectral peaks that satisfy the candidate selection criteria. One important observation is that only the 240 m range bin exhibits a clearly distinguishable atmospheric peak, while the remaining range bins contain numerous peaks of nearly equal amplitude. This illustrates the primary challenge addressed by the paper: under low SNR conditions, simply selecting the strongest peak can easily lead to an incorrect wind estimate because many clutter and noise peaks have comparable power.

To determine the best delay, the researchers repeat the mutual convolution process using eight different delay values, ranging from d_t to $8d_t$. The filtering results are presented in Figure 3.2. Each subfigure shows the Doppler spectra obtained using a different delay. From this we can conclude that the optimal delay is not same for every altitude.

Figure 3.2: Effect of delay in optimal solutions (marked by red dotted box) of different range bin after mutual convolution



To further analyze this behavior, the authors process 2500 radar frames collected on January 11, 2023. The atmosphere is divided into eight 40 m range intervals, and for every frame they record which delay produces the strongest mutual convolution result. These statistics are summarized in Table 3.1. Although different delays occasionally become optimal, the smallest delay d_t is selected far more frequently than the others across all range bins. The frequency of selecting larger delays decreases steadily. Based on these statistics, the authors conclude that while the optimal delay varies with atmospheric conditions, shorter delays generally provide the most reliable filtering performance and excessively large delays should be avoided because they reduce signal correlation.

Table 3.1: Statistics of optimal delay time for mutual convolution

Delay	RI	R2	R3	R4	R5	R6	R7	R8
d_t	1543	1503	1568	1471	1486	1488	1498	1568
$2d_t$	444	404	430	420	398	397	406	403
$3d_t$	179	203	195	224	214	218	223	196
$4d_t$	128	129	91	119	114	127	122	126
$5d_t$	70	72	68	73	85	80	78	70
$6d_t$	39	63	58	69	75	71	66	35
$7d_t$	42	60	38	63	59	51	48	47
$8d_t$	55	66	52	61	69	68	59	55

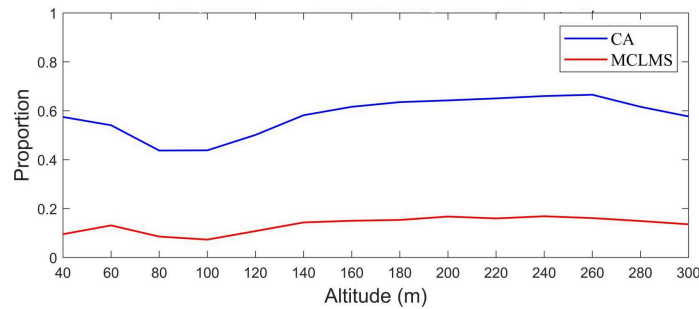
3.2 Peak Number and SNR

Here researchers have compared number of candidate spectral peaks and the SNR obtained from CA and proposed MCLMS filters. Ideally, each Doppler spectrum should contain only one dominant atmospheric peak. If many peaks remain, the peak-selection algorithm

becomes much more likely to choose an incorrect target. Using 5000 radar data sets, the researchers compare the conventional CA with the proposed MCLMS filter. The results are shown in Figure 3.3, which plots the proportion of spectra containing more than six candidate peaks. Across every altitude, the CA produces a high percentage of multi-peak spectra, typically between 45% and 70%. After applying the MCLMS filter, this proportion decreases dramatically to approximately 10–17%. This demonstrates that the adaptive filtering stage effectively suppresses clutter peaks while preserving the dominant atmospheric echo. The observations were recorded in following conditions:

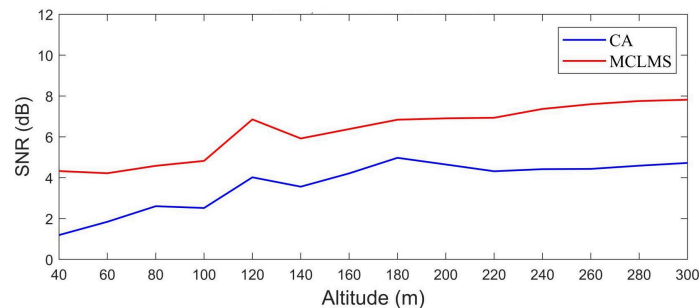
- **Time:** 6pm – 8pm (Dec 17, 2022)
- **Temperature:** $-2 \sim 9$ °C
- **Wind speed:** 4 ~ 8 m/s
- **Wind direction:** southeast

Figure 3.3: Proportion of the number exceeding 6 of candidate spectral peaks after applying the CA and the MCLMS filter respectively



The second performance metric is the SNR improvement, presented in Figure 3.4. Before filtering, the average SNR across all range bins remains below 5 dB, indicating very weak atmospheric echoes. After MCLMS filtering, the SNR increases by approximately 2–4 dB at every altitude. The higher SNR produces cleaner Doppler spectra and improves the reliability of subsequent peak selection.

Figure 3.4: SNR after applying the CA and the MCLMS filter respectively



3.3 Wind Profile Estimation Results

The researchers next evaluated the complete MCCF algorithm by comparing its wind estimates with three existing spectral estimation methods: AME, MPCF, and IME.

Figure 3.5 and Figure 3.6 present two radar observations recorded only one second apart. In Figure (a) each case shows the original Doppler spectra together with the wind profiles estimated by all four algorithms, while Figure (b) shows the spectra after MCLMS filtering together with the final MCCF estimates.

Figure 3.5: Comparison of four methods based on the data recorded on December 18, 2022 (15:19:32)

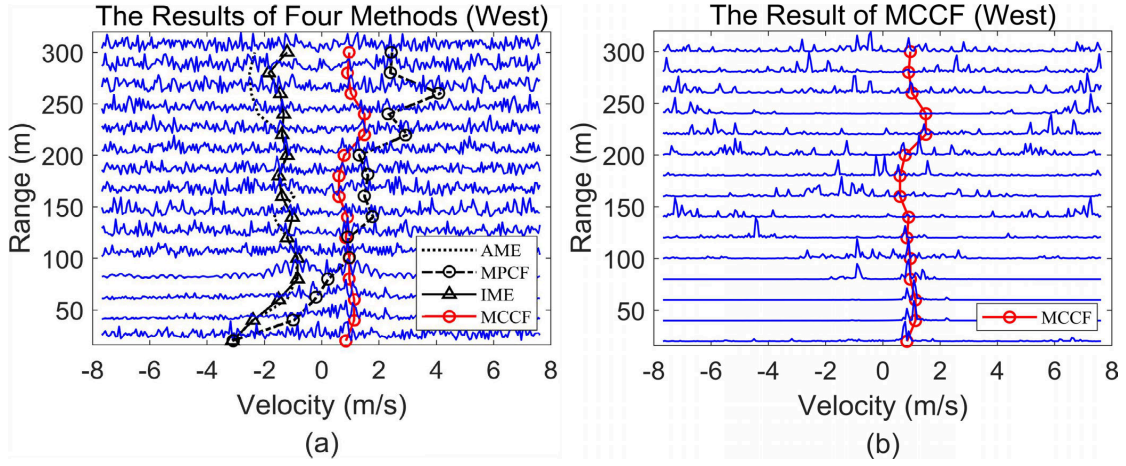
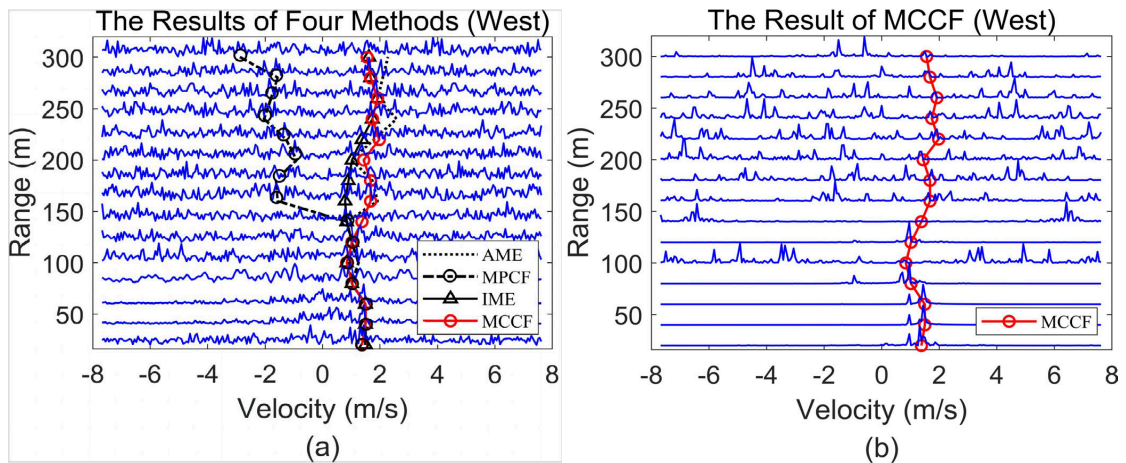


Figure 3.6: Comparison of four methods based on the data recorded on December 18, 2022 (15:19:33)



The researchers have observed that the AME and IME methods sometimes select clutter peaks in the lower range bins. Since subsequent peak selection depends on previous esti-

mates, these initial errors propagate upward, producing unstable wind profiles with large deviations. The MPCF method performs somewhat better but still exhibits considerable fluctuations, particularly above 100 m, because its candidate selection region is relatively broad. In contrast, the MCCF algorithm generates a much smoother and more continuous wind profile. The selected Doppler peaks follow a physically realistic gradual change in atmospheric wind velocity with altitude. Even though the two observations are separated by only one second, the other algorithms produce noticeably different wind profiles, whereas MCCF maintains nearly identical estimates, demonstrating superior temporal stability.

3.4 Scatter Plot and Correlation with LiDAR Observation

The final experiment validates the estimated wind velocities using an independent reference instrument – LiDAR. During March 3 to March 7, 2023, the VORTRAD radar and nearby LiDAR measured horizontal wind velocity. Because LiDAR generally has highest accuracy among all instruments, it provides a suitable reference for evaluating the radar algorithms.

For each method, the researchers compared 2500 pairs of radar and LiDAR measurements at an altitude of 200 m. The scatter plots shown in Figure 3.7 display radar-estimated velocity on the vertical axis and LiDAR velocity on the horizontal axis. The diagonal line represents perfect agreement between the two instruments.

The scatter plots clearly show that the MCCF estimates are much more tightly clustered around the diagonal than those produced by AME, MPCF, or IME. This indicates that the proposed algorithm produces wind velocity estimates much closer to the LiDAR measurements. The improvement is quantified using the correlation coefficient (CC), summarized in Table 3.2. Across all heights from 40 m to 280 m, the MCCF method consistently achieves correlation coefficients between 0.9576 and 0.9651, whereas the IME method reaches approximately 0.85–0.88, MPCF remains around 0.82–0.84, and AME performs worst, typically below 0.84. Since a correlation coefficient closer to unity indicates stronger agreement between two independent measurements, these results shows that the MCCF algorithm provides the most accurate wind velocity estimation among the four methods.

Figure 3.7: Scatter plot showing correlation among AME, MPCF, IME, MCCF methods against LiDAR. Color bars indicate data density

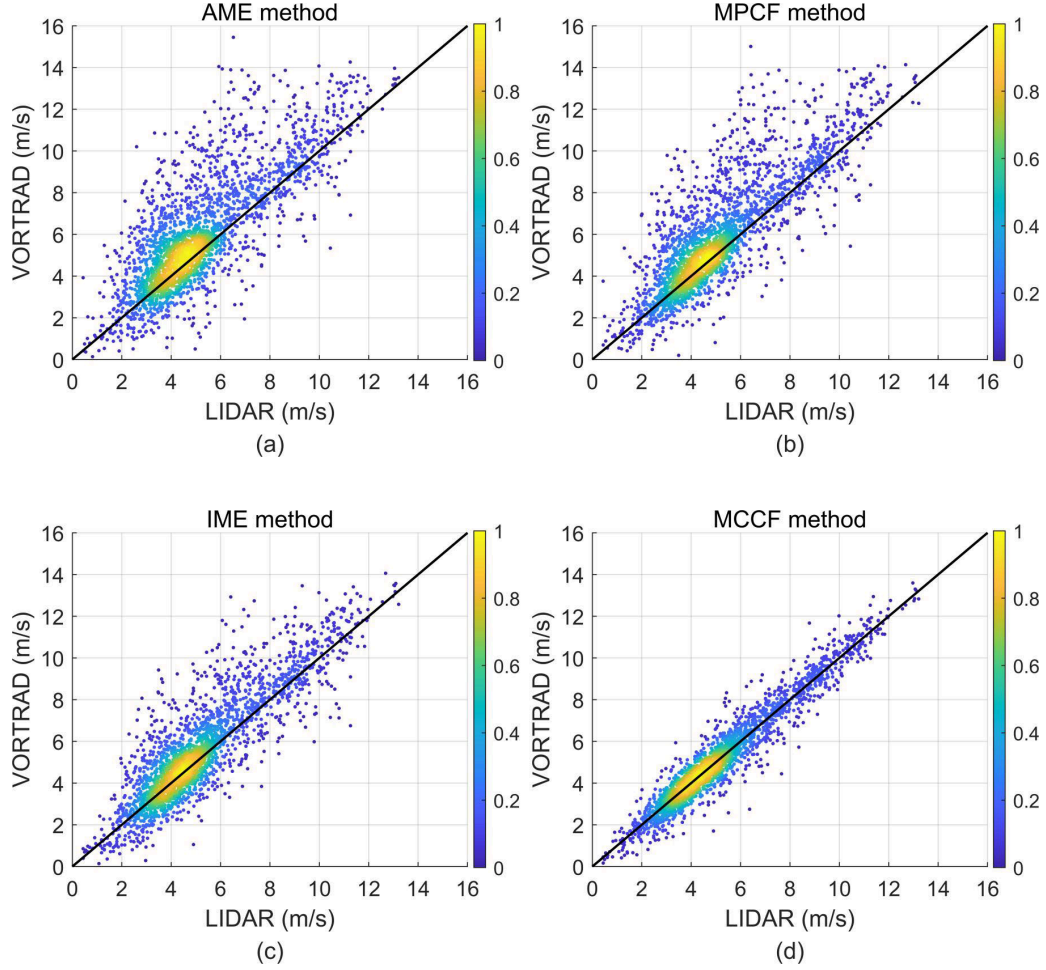


Table 3.2: Comparison of the correlation coefficient for VORTRAD and LiDAR data by using AME, MPCF, IME, MCCF

Height(m)	AME	MPCF	IME	MCCF
40	0.8417	0.8368	0.8369	0.9576
80	0.8319	0.8350	0.8461	0.9609
120	0.8320	0.8431	0.8695	0.9625
160	0.8137	0.8353	0.8761	0.9641
200	0.7908	0.8276	0.8764	0.9647
240	0.7830	0.8202	0.8738	0.9650
280	0.7837	0.8219	0.8744	0.9651

CHAPTER 4

CONCLUSION

Wind profile estimation using K-band Doppler radar is challenging under low SNR conditions because radar echoes are often contaminated by clutter, noise, and multiple Doppler peaks, making it difficult to identify the true atmospheric echo. To address this problem, a new method was proposed – Mutual Convolution Cost Function (MCCF) which combines the advantages of existing spectral estimation methods (AME, MPCF, and IME) with an improved CLMS filter. The proposed method first performs Mutual Convolution using multiple delayed versions of the radar signal to suppress uncorrelated noise while preserving coherent atmospheric echoes. An improved Mutual Convolution Complex Least Mean Square (MCLMS) filter is then applied to further reduce clutter and enhance the desired signal. Finally, candidate Doppler peaks are evaluated using a cost function that considers both spectral power and the physical continuity of wind velocity between adjacent range bins, enabling reliable selection of the atmospheric Doppler peak.

The proposed algorithm was evaluated using real atmospheric data collected by the VORTRAD K-band wind profiler. Experiments showed that the MCLMS filter reduced the number of false spectral peaks and improved the signal-to-noise ratio by approximately 2–4 dB. Compared with the AME, MPCF, and IME methods, the MCCF algorithm produced smoother and more continuous wind profiles while avoiding incorrect selection of clutter peaks. Validation against simultaneous LiDAR measurements demonstrated that MCCF achieved the highest correlation coefficient (approximately 0.96) across all tested altitudes, confirming its superior accuracy for wind profile estimation under low-SNR and multi-peak conditions. The authors note that the current method is validated only for clear-weather conditions and does not account for precipitation echoes such as rain or snow, leaving this as future work.

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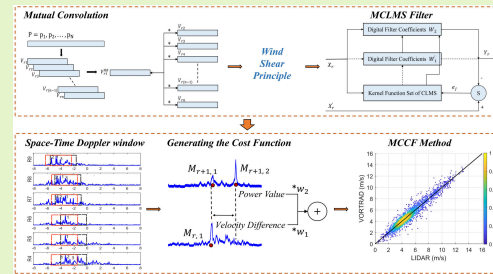
Appendix - 1: Reference Paper

Wind Profile Estimation for K-Band Doppler Radar Based on Mutual Convolution Cost Function Method

Zaixing Tan¹, Guangli Yang¹, Siye Wang, Lijun Wang, Yvxin Fan¹,
Rui Wang¹, Senior Member, IEEE, and Yong Luo¹, Member, IEEE

Abstract—At present, K-band Doppler radar has been proposed for detecting the motion state of atmospheric turbulence and obtaining wind velocity and direction. However, the echo signal of the turbulence is often interfered with strong noise, and there are multiple spectral peaks in the power spectrum, which greatly affect the overall detection accuracy of the radar. Traditional wind profile estimation methods directly analyze the coherent accumulation (CA) signal using a cost function, which is susceptible to clutter and noise during the selection of the candidate spectral peaks. To overcome the shortcomings of traditional methods under the condition of low signal-to-noise ratio (SNR) and multiple peaks, in this article, we propose a novel method based on the mutual convolution cost function (MCCF), which utilizes the mutual convolution and the complex least mean square (MCLMS) method to filter the CA signal, and then generates a cost function to select the optimal spectral peaks in the space-time Doppler window. This systematically solves the problem of turbulence echoes being interfered with strong noise and multiple peaks. Finally, the commonly used methods are compared with the proposed method, and the statistical results of these methods are verified by using LIDAR data. The results indicate that the proposed method can improve the SNR by 2–4 dB and improve the overall correlation coefficient (CC) by about 10% of the radar.

Index Terms—Low signal-to-noise ratio (SNR), multiple peaks, mutual convolution and the complex least mean square (MCLMS), mutual convolution cost function (MCCF), wind profile estimation.



I. INTRODUCTION

IN RECENT years, wind power generation has become one of the important ways to provide clean energy. Numerous scientists have dedicated substantial efforts to research in the field of wind energy and made significant achievements [1], [2], [3]. Within the realm of wind energy detection, LIDAR and meteorological masts provide schemes for accurately detecting wind velocity and direction [4], [5]. A wind turbine

adjusts its angle in response to the wind direction to enhance power generation efficiency [6], [7]. During the research, we identified that K-band Doppler radar can also effectively detect near-surface atmospheric turbulence, offering a novel solution for wind energy detection [8].

The detection of atmospheric turbulence poses numerous challenges for K-band Doppler radar, mainly reflected in the following aspects.

- 1) Atmospheric turbulence characteristics vary depending on factors such as season, temperature, humidity, and time.
- 2) In colder environments, a decrease in density significantly lowers the signal-to-noise ratio (SNR) of radar echoes, causing difficulties in extracting wind information from noise.
- 3) Due to the circular detection area of the antenna, echoes often contain multiple spectral peaks, making it difficult to accurately determine the specific frequency offset of the target [9], [10], [11]. Therefore, our objective is to propose a novel methodology focusing on enhancing SNR and mitigating the presence of multiple spectral peaks in echo.

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In the data processing algorithm of wind profile radar, spectral estimation is primarily conducted in the frequency domain by most researchers. The traditional estimation method assumes that the target signal is the most prominent spectral peak on the spectrum axis [12]. However, the accuracy of this estimation method can be significantly compromised in the presence of strong noise. Therefore, Clothiaux et al. [13] proposed identifying prospective spectral peaks by selecting the longest chain and the maximum sum of power spectra values as the principle [13]. Anandan et al. [14] proposed a method called adaptive moment estimation (AME) based on SNR and wind shear. Sinha et al. [15], [16] proposed the multiparameter cost function (MPCF) method for estimating the wind profile. Li et al. [17] proposed an improved moment estimation (IME) method for wind estimation. Bhatta et al. [18] proposed the Viterbi data association (VDA) method to track the wind profile. These methods have demonstrated favorable results in numerous experiments, particularly when the SNR exceeds 5 dB. However, their performance on the *K*-band radar may deteriorate when the SNR is lower and multiple spectral peaks are present, thereby impacting accuracy.

In the field of turbulence detection, the traditional denoising method is coherent accumulation (CA), which exploits the instability of noise to reduce its power [19]. However, where both turbulence density and atmospheric temperature are very low, the performance of CA is subpar. The complex least mean square (CLMS) filter is proposed to adaptively filter the radar echo signal in the complex domain. Later, Took and Mandic [20] proposed the quaternion least mean square (QLMS) for wind vector evaluation based on CLMS. However, the CLMS filter has been proven that its filtering performance will reduce unstable signals [21]. To improve the filtering performance of adaptive filtering for unstable signals, Pei and Ding [22] proposed the fractional Fourier transform (FRFT) method to make the energy of unstable signals more concentrated. Chen et al. [23] proposed the normalized leakage LMS (NL-LMS) to maximize the energy of the target signal and suppress clutter in the FRFT domain. Shi et al. [24] proposed shrinkage widely linear CLMS (SWL-CLMS), which significantly improved the convergence speed and reduced the steady-state misalignment. Menguc and Acir [25] proposed the augmented complex valued least mean kurtosis (ACLMK) filter, which utilizes augmented statistics to define the negative kurtosis of the complex error signal as a cost function. Furthermore, Zhang et al. [21] proposed a wide linear complex estimation input adaptive filter (WLC-EIAF), designed to achieve unbiased estimation in scenarios where input and output signals are affected by noise interference.

In response to the challenges faced by the *K*-band wind radar, the main innovation of this article is to obtain a space-time Doppler window, based on spectrum estimation results in the previous range bin, as well as the spectrum estimation results of same the range bin in the previous frame of data, and the range of wind shear. The space-time Doppler window is utilized to constrain the detection range of mutual convolution, obtaining the input signal and desired signal for the mutual convolution and the CLMS (MCLMS) filter, and the CLMS filter structure has been improved. The

TABLE I
SPECIFICATIONS RELEVANT TO THE VORTRAD

Number	Specification	Value
1	Frequency band	K-band
2	Operation mode	Single frequency pulse
3	Number of beams	4 (East, South, West, North)
4	Tilt Angle	15°
5	Pulse period	2 us
6	Pulse width	≈ 133.3 ns
7	No. of coherent accumulation	50
8	Slow sampling points	2048
9	FFT points	1024
10	Maximum range	300 m
11	Window coefficient	2
12	Maximum radial velocity	30 m/s
13	Radial velocity resolution	0.06 m/s
14	Range resolution	20 m

cost function is ultimately employed for spectrum estimation within the results of MCLMS filtering, which is known as the mutual convolution cost function (MCCF) algorithm. Before implementing the MCCF, it is necessary to preprocess the echo signal by preliminarily improving the SNR and removing direct current (dc) components and ground clutter.

The rest of this article is organized as follows. Section II introduces the radar system and the echo signal of four beams. Section III introduces the process of the MCCF algorithm in detail. Section IV introduces the performance of the proposed method, compares it with the previous algorithms, and validates it using data from the same period of the LIDAR. Finally, in Section V, we summarize the entire article.

II. EQUIPMENT DESCRIPTION

This article utilizes the *K*-band vortex radar (VORTRAD) as experiment equipment provided by Suzhou Dufeng Technology Company, which estimates wind profile by detecting Doppler frequency offset generated by atmospheric turbulence. The VORTRAD system is mainly composed of four parts, transmission link, reception link, baseband circuit, and host, the important specifications are summarized in Table I.

As shown in Fig. 1, the experimental environment consists of VORTRAD with a shield and LIDAR to minimize errors arising from environmental changes, two devices are placed near the wind turbine with an interval distance of about 3 m. The distance resolution, velocity resolution, and measurement distance of the two devices are consistent, and the accuracy of this LIDAR under clear sky conditions reaches 99%.

III. RESEARCH METHODS

This section introduces the data processing algorithm of turbulence echo in detail and mainly addresses the issues of low SNR and multiple spectral peaks by the MCCF algorithm. The algorithm is divided into five parts, signal preprocessing, setting the space-time Doppler window, mutual convolution, MCLMS filter, and generating the cost function. In order



Fig. 1. Test environment VORTRAD radar and LIDAR were built at Changxing Island Wind Farm in Shanghai, China to compare four methods according to the turbulence echo data.

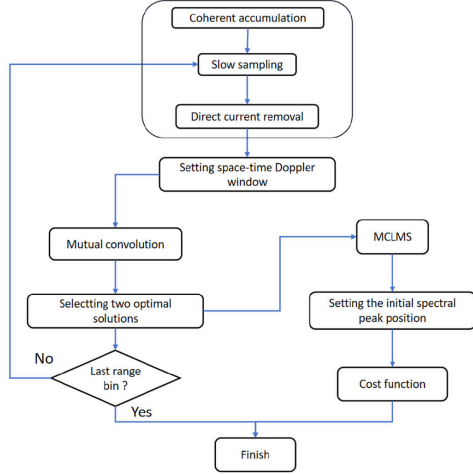


Fig. 2. Execution process of the MCCF algorithm which limits the detection range of mutual convolution by utilizing the space-time Doppler window and subsequently employs the cost function to estimate the spectrum within the filtering results of MCLMS.

to show completeness, the MCCF algorithm begins with the discrete signal sampled by the ADC. The specific workflow is shown in Fig. 2.

A. Signal Preprocessing

Turbulence is characterized as a weak target. Over a short period, turbulence moves at a uniform velocity, rendering its echo akin to a stable periodic signal. Conversely, noise exhibits instability and lacks periodicity. Hence, CA of the echo signal over a short period, followed by averaging, serves to amplify the signal power and diminish the noise power. Here, the result of the first CA is x_{c1} , and the 2-D matrix generated by the result after N -order CA can be given as

$$X = [x_{c1}^T, x_{c2}^T, \dots, x_{c(N-1)}^T, x_{cN}^T] \quad (1)$$

where T represents the transposition of a vector.

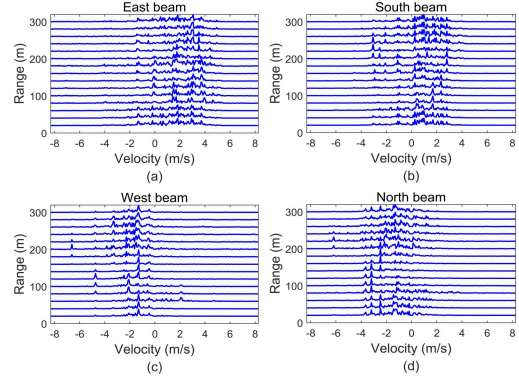


Fig. 3. (a)–(d) Represent the range Doppler power spectrum of the four beams respectively, which include low SNR and multiple peaks echo, was recorded from wind farm on January 11, 2023.

The column vector of the X represents the result of one accumulation. To detect the motion state of turbulence at different altitudes, we need to extract data from a specific range bin. Therefore, it is necessary to extract a row vector from X to generate a new vector. This process is called slow sampling in the distance dimension. The slow sampling frequency must satisfy the Nyquist criterion in order to avoid blurring the maximum turbulence velocity, and the slow sampling frequency f_{dmin} can be given as

$$f_{dmin} \geq \frac{4v_{max}}{\lambda} \quad (2)$$

where v_{max} represents the maximum velocity range and λ represents the work wavelength of the radar.

The VORTRAD adopts a zero-intermediate-frequency RF transceiver architecture, leading to the echo signal containing a large dc power, and ground clutter with a small frequency offset, which will significantly impact the subsequent denoising algorithm. Therefore, replacing five points around the zero frequency with the noise spectral peak can eliminate the zero frequency and most ground clutter without affecting the spectral peak of turbulence. After removing the dc and ground clutter, convert the row vector into a time domain signal.

The preprocessed echo data of the VORTRAD are shown in Fig. 3, it can be seen that only the echo signal of the west beam exhibits good SNR, while the other three beams have low SNR with multiple spectral peaks. At the first range bin of the north beam and east beam, there are multiple spectral peaks with similar power, making it difficult to determine the initial spectral peak of the cost function.

B. Setting the Space-Time Doppler Window

As pointed out in [17], the turbulent motion state between adjacent altitudes continuously changes, and wind shear is associated with a difference in altitude. Furthermore, wind shear is typically less than 0.5 m/s in the range of 10 m and is less than 1 m/s in the range of 20 m. Generally,

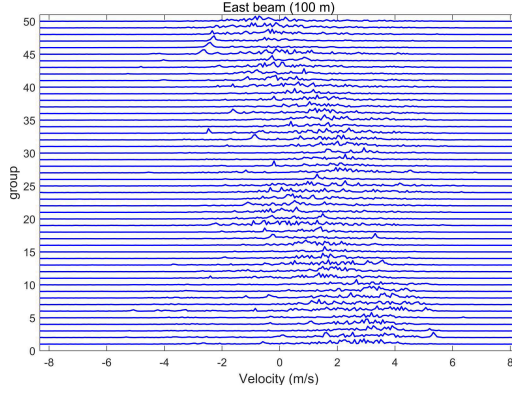


Fig. 4. Doppler power spectra of the east beam of 100 m were recorded on January 11, 2023 in Changxing Island, China.

researchers believe that the signal's variation range typically falls within 20% total bandwidth of the coherent filter, only candidate spectral peaks within this bandwidth need to be selected. However, when the frequency of slow sampling is fixed, additional slow sampling points are required to increase the velocity resolution. Therefore, it is necessary to adjust the size of the Doppler window based on the desired velocity resolution.

Furthermore, wind velocity exhibits continuity not only across different altitudes but also over time at the same altitude [5], as shown in Fig. 4, which illustrates the echoes of 50 consecutive groups of preprocessed data at 100 m in the east beam, it can be seen that over some time, wind velocity exhibits significant continuity at 100 m.

Therefore, when setting the Doppler window, it is essential to consider both spatial and temporal aspects.

First, considering space continuity, assuming that the spectrum estimation result of the previous range bin is F_0 , the space Doppler window can be represented as

$$W_1 = (F_0 - \alpha d_f, F_0 + \alpha d_f) \quad (3)$$

where d_f is the frequency offset corresponding to wind shear and α is the window coefficient related to altitude interval.

The value of d_f is related to the velocity resolution of the radar system. If the maximum value of wind shear is V_d , then d_f can be expressed as

$$d_f = \frac{V_d}{d_v} = \frac{V_d}{\frac{f_{dmin} \lambda}{L}} \quad (4)$$

where d_v indicates velocity resolution and L indicates the points of FFT.

Then, considering time continuity, assuming the previous spectrum estimation result at the same range bin is T_0 , the time Doppler window can be represented as

$$W_2 = (T_0 - \alpha d_f, T_0 + \alpha d_f). \quad (5)$$

Therefore, the estimation window for spectrum estimation in the next range bin is determined by the intersection of

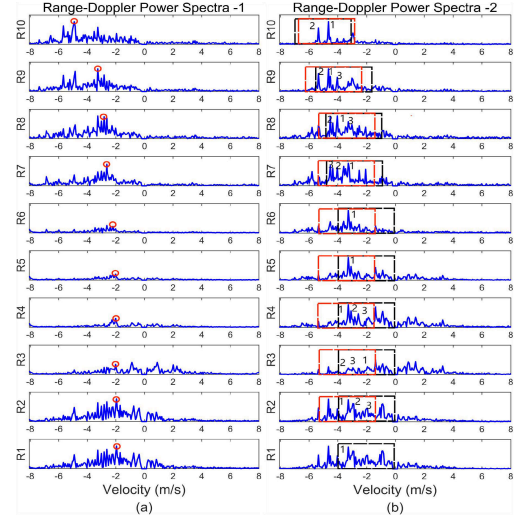


Fig. 5. (a) Echo from the previous moment with spectral peak selection results are marked by red circles. (b) Echo from the following moment is marked with the space-time Doppler window. Red rectangle represents space Doppler window, the black rectangle represents time Doppler window, and candidate peaks are marked by number.

two Doppler windows. However, when the intersection of two windows is less than αd_f , the space Doppler window is still utilized as a selection range. Abbreviated it as the space-time Doppler window, which can be expressed as

$$ST - W = \begin{cases} W_1 \cap W_2, & \text{if } (W_1 \cap W_2) \geq \alpha d_f \\ W_1, & \text{else.} \end{cases} \quad (6)$$

As shown in Fig. 5(a), the echo from the previous moment is visualized, with spectral peak selection results marked by red circles. In Fig. 5(b), the echo from the following moment is marked with the space-time Doppler window.

C. Mutual Convolution

To minimize the error rate of spectral estimation, efforts should be made to attenuate noise and retain significant spectral peaks before identifying candidate spectral peaks.

First, take a row vector $P = (p_1, p_2, \dots, p_N)$ of one range bin from the preprocessing data. Then, take out vectors with equal length from P to form a new row vector which can be given as

$$V_{r1} = (p_1, p_2, \dots, p_{\frac{N}{2}}). \quad (7)$$

The second vector is formed by extracting delay data, and the number of delay points can be expressed as I , based on the total length of the vector and the desired number of extracted vectors, it can be expressed as

$$I = \frac{N}{2(n-1)} \quad (8)$$

where N represents the total length of the vector P , n represents the desired number of extracted vectors, and a new

matrix V can be given as

$$V = [V_{r1}, V_{r2}, \dots, V_{rn}]. \quad (9)$$

Since it cannot determine whether the noise and clutter of two arbitrarily selected signal segments have maximum uncorrelated characteristics, we make full use of the principle that stable signal frequency remains constant for a period while the noise of two periods is not correlated. The initial vector and the delay vector are taken from V as the input signal of mutual convolution, and then the power spectrum value of mutual convolution is calculated respectively, and the best result is selected based on wind shear; the mutual convolution process can be given as

$$\begin{cases} V_{C1} = V_{r2} * V_{r1}^{\text{HR}} \\ V_{C2} = V_{r3} * V_{r1}^{\text{HR}} \\ \vdots \\ V_{C(n-1)} = V_{rn} * V_{r1}^{\text{HR}} \end{cases} \quad (10)$$

where HR means that the conjugate and then reverse order.

After the above calculation, the number of convolution results is $n-1$ and the length of each calculation result becomes $N-1$. To maintain the same time complexity and velocity resolution of calculation, a new matrix can be generated by extracting data from the middle of the vector, with the number being $N/2$, it can be given as

$$\begin{cases} V'_{C1} = V_{C1} \left(\frac{N}{4} : \frac{3N}{4} \right) \\ V'_{C2} = V_{C2} \left(\frac{N}{4} : \frac{3N}{4} \right) \\ \vdots \\ V'_{C(n-1)} = V_{C(n-1)} \left(\frac{N}{4} : \frac{3N}{4} \right) \end{cases} \iff [V'_c]. \quad (11)$$

Next, the power spectrum of each set of row vectors in V'_c is calculated, and the maximum value in the ST-W of each set of power spectrum data is identified to obtain an array, it can be given as

$$V'_{cf} = (\max \{ |\mathcal{F}([V'_c])| \}) \quad (12)$$

where $|\mathcal{F}()$ represents the Fourier transform calculated first before the modulus value is calculated.

In the generated maximum power spectrum value array, comparing the values in the array V'_{cf} , record convolution calculation results corresponding to the maximum power spectrum value as the optimal solution X_r .

Then, convolution results corresponding to adjacent delay times are recorded as suboptimal solutions X'_r , and in the MCCF algorithm, values adjacent to the left of the maximum value are uniformly selected. If the maximum value in (12) is the first value, an adjacent value is selected to the right of the maximum value.

D. MCLMS

Taking X_r and X'_r as input signal and desired signal of the MCLMS adaptive filter, the clutter will be further suppressed. Unlike the previous CLMS filter, this one transfers the filter

coefficients as parameters. After filter coefficients have been fully updated, the recognition ability of stable signals becomes more prominent, and it excels at solving the problem of multiple strong clutter interference. Its specific calculation process is as follows.

- 1) The optimal solution is represented as an $N/2 \times N/2$ -dimensional row vector and the length of the coefficient filter is denoted as k . Calculating the normalized parameters of filter coefficients as

$$u = \frac{1}{\sum_{j=1}^{N/2} |X_r(j)|^2}. \quad (13)$$

- 2) Taking out data from X_r in turn, record as

$$X = \begin{cases} [\text{zeros}(1, k-j), X_r(1:j)], & (j < k) \\ X_r(j-k+1:j), & (j \geq k) \end{cases} \quad j \in [1, N/2]. \quad (14)$$

When the data taken out are not equal to the filter coefficients, 0 must be filled in front of the data. Otherwise, dc components will easily appear in filtering results.

- 3) Initializing filter coefficients as $W_1 = 0$, setting the filter kernel function as

$$\begin{cases} Y_j = W_1 \cdot X^T \\ e_j = X'_{rj} - Y_j \\ W'_1 = 0.98W_1 + u \cdot e_j \cdot X^H \end{cases} \quad (15)$$

Repeating the above steps until all data are filtered, the coefficients have been fully updated, but the filtering results are not outputted at this time. We will reinitialize filter coefficients to obtain the new coefficients as

$$W_2 = W'_1 \quad (16)$$

- 4) Then, using the new filter coefficients, reexecuting steps (b) and (c) again to obtain the filtering result, and record it as Y_r .

E. Generating the Cost Function

When employing a cost function for spectral peak selection, it is typically assumed that there exists only one prominent spectral peak in the first range bin. However, the VORTRAD system often displays multiple peaks in the first range bin. Therefore, when selecting spectral peaks in the first range bin, the first step is to calculate the average value corresponding to the power spectrum of the convolution network, it can be given as

$$M_1 = \left(\sum_{i=1}^{n-1} \mathcal{F}(V'_{Ci}) \right) / (n-1). \quad (17)$$

The most prominent spectral peak is selected as the result of spectral estimation in the first range bin, which can be given as

$$M_{1,1} = \max |M_1|. \quad (18)$$

The power spectrum of MCLMS filtering results is used for peak selection, which can be given as

$$M_r = |\mathcal{F}(Y_r)|. \quad (19)$$

Then the spectral peaks in each ST-W are sorted according to their power, and up to three spectral peaks are selected in each ST-W.

Supposing maximum spectral peak within the ST-W as $M_{r\max}$ and candidate spectral peaks are selected as shown in Fig. 5(b), the candidate spectral peaks can be given as

$$M_{r,j} \geq 0.4M_{r\max} \quad (20)$$

where $M_{r,j}$ represents the power value corresponding to the j th spectral peak of the r th range bin.

According to the selected candidate spectral peak, the cost value corresponding to the calculated velocity difference of the spectral peak can be expressed as

$$V_{r+1,j} = 1 - \left| \frac{(M_{r+1} - M_r) \cdot d_v}{\alpha V_d} \right| \quad (21)$$

where d_v represents the velocity resolution, V_d represents the maximum wind shear, α is the window coefficient, and $V_{r+1,j}$ represents the velocity cost value corresponding to the j th spectral peak of the $(r+1)$ th range bin.

Next, by considering the velocity cost value and power value of candidate spectral peaks as two constraints in the cost function, the MCCF cost function can be given as

$$F_{r+1,j} = w_1 V_{r+1,j} + w_2 M_{r+1,j} \quad \sum_{i=1}^2 w_i = 1 \quad (22)$$

where w_1 and w_2 represent the respective weights of the two constraints, and their specific configurations can be referred to in [15], $F_{r+1,j}$ represents the cost value from the spectral peak $M_{r,j}$ to $M_{r+1,j}$. Finally, the spectral peak corresponding to the maximum cost value in each range bin is the expected spectral peak.

IV. EXPERIMENT AND RESULTS

This section demonstrates the effectiveness of the MCCF method by presenting experimental results. The first part verifies the performance of different delay times. The second part focuses on evaluating the performance of the MCCF, quantifying the number of spectral peaks and the SNR after filtering. In the third part, the estimation results of the MCCF algorithm are obtained and compared with the spectral estimation results of the AME, MPCF, and IME algorithms. The fourth part compares the horizontal wind velocity obtained from different algorithms with the LIDAR data and calculates the correlation coefficients (CCs) of the four methods.

A. Performance Evaluation at Different Delay Times

The echo signal SNR remains very low after preprocessing, as shown in Fig. 6, where the signal represents the echo spectra from 220 to 300 m after preprocessing on Changxing Island on December 19, 2022. Among them, the Doppler frequency offset that may be caused by turbulent motion is marked with red dots; it can be seen that there is a single prominent spectral peak at 240 m with little interference from other spectral peaks and there are multiple peaks present in other range bins; and the power of each peak is approximately equal and is significantly disturbed by noise. In the traditional

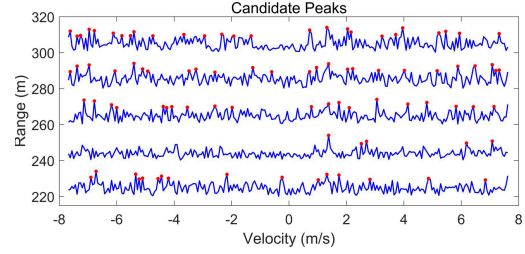


Fig. 6. Range-Doppler power spectra recorded on December 20, 2022 (15:22:19) and the candidate peaks are marked by red dots.

spectral estimation algorithm, the peaks marked by red dots are set as candidate peaks, and an optimal solution is derived by evaluating a cost function. However, we cannot simply consider the peak with a high-power value as a potential target, the marked peaks may be the noise. Based on the wind shear principle, the error rate of the spectral estimation algorithm will increase.

Therefore, we hope to remove the interference caused by clutter and noise before selecting candidate spectral peaks. The matched filter is usually used to achieve the best reception effect, but the specific delay time of the matched filter is not given to constrain convolution calculation; in the MCCF method, we calculate the convolution of the delayed signal and initial signal based on (10) and observe that denoising effect varies with different delay times. The convolution results of the signal in Fig. 6 are shown in Fig. 7, with an increasing delay time from (a) to (h), setting the minimum delay as d_t , and selecting the best filtering result based on (12) for each range bin marked by the red dotted box; it can be seen that the optimal delay corresponding to the signal of 220 m is $8d_t$, the optimal delay corresponding to 240 m is $4d_t$, the optimal delay corresponding to 260 m is $6d_t$, the optimal delay corresponding to 280 m is $3d_t$, and the optimal delay corresponding to 300 m is d_t .

To find the relationship between the optimal filtering result and the delay time, frame data are divided into eight range bins at intervals of 40 m, a total of 2500 frames of echo data from Changxing Island on January 11, 2023 were processed, and the number of different optimal delay times was recorded. As shown in Table II, it can be concluded that the optimal delay time is concentrated at d_t . As the delay time increases, the number of statistics gradually decreases, but it is still impossible to determine a specific optimal delay time. Moreover, the delay time cannot be increased indefinitely in the signal processing based on CA.

B. Comparison of Peak Number and SNR

The data from Changxing Island, China, comprising 5000 sets, were processed to calculate the number of candidate spectral peaks using the CA and MCLMS filter. Fig. 8 exhibits the proportion of the number exceeding 6 of candidate spectral peaks within the space-time Doppler window, which is evident that the proportion of multi-peaks has significantly decreased after filtering, indicating that the MCLMS filter effectively

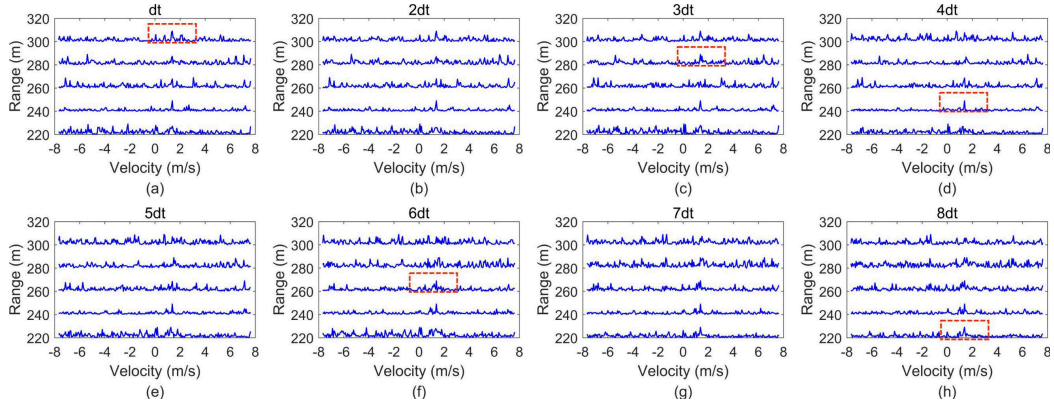


Fig. 7. Optimal solutions are marked by red dotted box of different range bin after mutual convolution for Fig. 6. (a)–(h) Represent the filtering effect of d_t – $8d_t$ delay time respectively.

TABLE II
STATISTICS OF OPTIMAL DELAY TIME FOR MUTUAL CONVOLUTION

Delay	R1	R2	R3	R4	R5	R6	R7	R8
d_t	1543	1503	1568	1471	1486	1488	1498	1568
$2d_t$	444	404	430	420	398	397	406	403
$3d_t$	179	203	195	224	214	218	223	196
$4d_t$	128	129	91	119	114	127	122	126
$5d_t$	70	72	68	73	85	80	78	70
$6d_t$	39	63	58	69	75	71	66	35
$7d_t$	42	60	38	63	59	51	48	47
$8d_t$	55	66	52	61	69	68	59	55

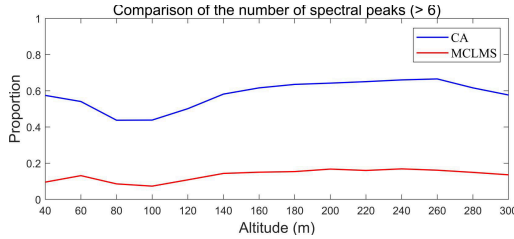


Fig. 8. Proportion of the number exceeding 6 of candidate spectral peaks after applying the CA and the MCLMS filter respectively.

TABLE III
TESTING ENVIRONMENT OF THE SYSTEM

Time	T	Wind direction	Wind speed
2022.12.19 18:00-20:00	-2 ~ 9°C	southeast	4 ~ 8 m/s

mitigates the occurrence of multi-peaks in turbulent echoes. The system's test environment is presented in Table III, where lower nighttime temperatures are more suitable for evaluating system performance. Fig. 9 shows that the average SNR in each range bin was generally below 5 dB before filtering. However, the SNR of the echoes increased by 2–4 dB after filtering.

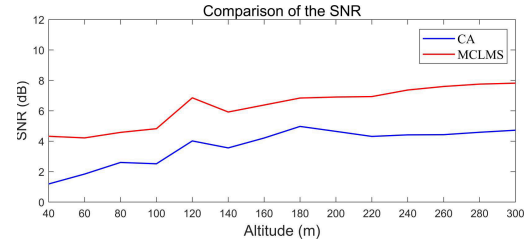


Fig. 9. Compare the SNR after applying the CA and the MCLMS filter respectively.

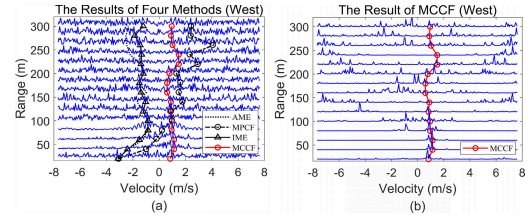


Fig. 10. Comparison of four methods (West) based on the data recorded on December 18, 2022 (15:19:32). (a) Displays the Doppler spectrum of the VORTRAD west beam before applying the MCCF method and the results of four methods. (b) Displays the Doppler spectrum and result of the VORTRAD west beam after applying the MCCF method.

C. Comparison of Wind Profile Estimation Results

Fig. 10(a) displays the Doppler spectrum of the VORTRAD west beam before applying the MCCF method on December 18, 2022 (15:19:32) and the results of four methods. The minimum SNR is 1.48 dB, the maximum SNR is 8.65 dB, and the average SNR is 3.45 dB. The Doppler spectrum after applying the MCLMS filter and the estimation result of the MCCF method are displayed in Fig. 10(b), it is apparent that there is a notable difference in the Doppler velocity estimated by the four methods. The AME and the IME methods exhibit relatively minor velocity fluctuations, but an unstable clutter was selected in the first range bin, leading to a substantial

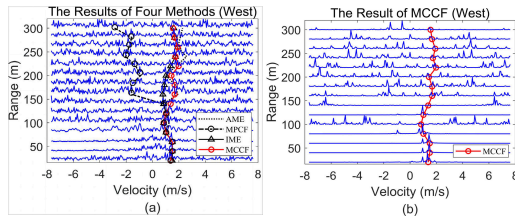


Fig. 11. Comparison of four methods is based on the data recorded on December 18, 2022 (15:19:33). (a) Displays the Doppler spectrum of the VORTRAD west beam before applying the MCCF method and the results of four methods. (b) Displays the Doppler spectrum and result of the VORTRAD west beam after applying the MCCF method.

overall deviation in subsequent velocity estimates. In contrast, the MPCF has a wider selection range in choosing candidate spectral peaks, resulting in greater velocity fluctuations above 100 m. From Fig. 10(b), it is evident that the stable signal of the first range bin is approximately 1 m/s. Therefore, the MCCF method has better continuity in selecting spectral peaks and the selected signals are more stable, avoiding selecting high-power clutter as much as possible.

Fig. 11(a) displays the Doppler spectrum of the VORTRAD west beam before applying the MCCF method on December 18, 2022 (15:19:33) and the results of four methods. The minimum SNR is 1.51 dB, the maximum SNR is 9.80 dB, and the average SNR is 3.29 dB. Fig. 11(b) displays the Doppler spectrum after applying the MCLMS filter and the estimation results of the MCCF method. The time difference between the echoes in Figs. 10 and 11 is merely one second. In general, the wind profile estimation results of the echoes in the same beam do not undergo significant changes over a short period. However, there are significant differences in the estimation results of the AME, MPCF, and IME methods. These differences become even more pronounced, leading to worsened continuity of wind velocity. The wind velocity flow pattern of echoes from the same beam is similar, and there is no abrupt velocity change within one second based on the estimation results of MCCF, which is more consistent with the wind shear principle and result is more accurate.

D. Scatter Plot and Correlation Coefficient of Wind Velocity Extraction

As shown in Fig. 1, VORTRAD and LIDAR are positioned close, with a considerable distance separating the two pieces of equipment from the turbine to prevent interference. From March 3, 2023 to March 7, 2023, two pieces of equipment were tested synchronously, and the echo data from VORTRAD were processed by four methods respectively. The obtained horizontal wind velocity was compared with the horizontal wind velocity measured by the LIDAR at the corresponding time, and a scatter plot was generated. During this period, temperatures ranged from 4 °C to 15 °C, with the average SNR of the VORTRAD echo remaining below 5 dB.

Scatter plots were generated by selecting 2500 points, with color bars indicating data density. The comparison results of 200 m are shown in Fig. 12. In the case of low SNR, the IME method exhibits certain improvements over the AME and MPCF methods. The velocity extracted by the IME method

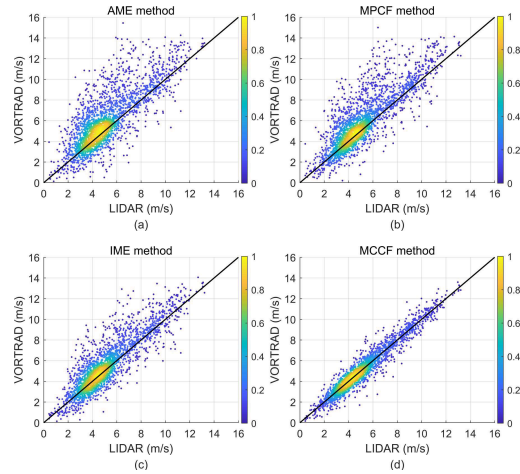


Fig. 12. Color bars indicate data density and show the comparison of horizontal wind velocity from the VORTRAD and LIDAR at the height of 200 m. (a) Comparison result of the AME method. (b) Comparison result of the MPCF method. (c) Comparison result of the IME method. (d) Comparison result of the MCCF method.

TABLE IV
COMPARISON OF THE CORRELATION COEFFICIENT FOR VORTRAD AND LIDAR DATA BY USING AME, MPCF, IME, MCCF

Height(m)	AME	MPCF	IME	MCCF
40	0.8417	0.8368	0.8369	0.9576
80	0.8319	0.8350	0.8461	0.9609
120	0.8320	0.8431	0.8695	0.9625
160	0.8137	0.8353	0.8761	0.9641
200	0.7908	0.8276	0.8764	0.9647
240	0.7830	0.8202	0.8738	0.9650
280	0.7837	0.8219	0.8744	0.9651

aligns more closely with the LIDAR velocity. On the other hand, the velocities extracted by the MPCF and AME methods are more divergent. At the same time, the horizontal velocity estimated by the MCCF method aligns noticeably well with the LIDAR, exhibiting a more pronounced velocity correlation, better than the other three methods. Based on the comparison results of 200 m, it can be inferred that only the horizontal velocity of the MCCF method closely aligns with the LIDAR, while the other three methods are significantly different. This indicates that the MCCF method has better processing ability for echoes with low SNR and multiple spectral peaks.

The results are statistically analyzed, as shown in Table IV, the CC of the horizontal wind velocity extracted by four methods at different range bins are compared. A higher value of CC indicates a greater fit degree between the two sets of data. Since the working frequency of VORTRAD is in the K band, the target detected by the antenna is a large area [8], while the detection area of LIDAR is relatively smaller, the information received by the two systems has certain differences, which is the main source of error. However, the CC value of the MCCF algorithm can still reach 0.95 after processing the data, while the other three methods lower than 0.9, indicating that the

horizontal wind velocity extracted by the MCCF algorithm is more accurate than the other three methods.

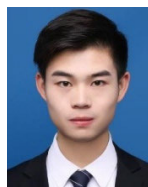
V. CONCLUSION

In this article, we introduce a novel K -band wind profile radar, including its measurement principle, specifications, and a novel data processing method. This method combines the advantages of AME, MPCF, IME methods, and CLMS filtering in processing wind profile radar echo data. In the MCCF algorithm, the impact of different delay times on the performance of convolution computation was first analyzed. Then, the continuous characteristics of velocity resolution, wind shear, and turbulence in both time and space were utilized to adaptively adjust the size of the Doppler window.

Next, the structure of the CLMS filter was improved, and the convolution network was constrained by a space-time Doppler window to obtain the input signal and desired signal of the MCLMS, and the cost function was used to select the optimal solution for each range bin. Finally, experiments have demonstrated that the MCCF method can accurately extract wind profile and velocity information under the condition of low SNR and multiple peaks. However, the MCCF method only considers sunny days, assuming that the received signal only includes turbulence signals, clutter signals, and interference signals, without considering other signals caused by rain and snow weather, so further improvement is needed.

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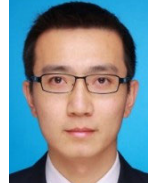
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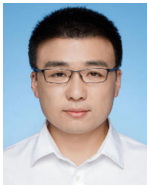
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APPENDIX - 2: PRESENTATION SLIDES

1. Introduction ▲

1.1 Background

- Wind profiles are estimated from Doppler Radars via signal processing techniques.
- Radar echoes are often heavily corrupted by background noise (gets worse in cold climate)
- Traditional methods (like coherent accumulation) works well only when SNR is high
- Under low SNR conditions, multiple peaks are observed

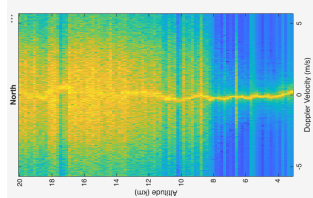


Figure 1: Doppler spectrum

4

1. Introduction ▲

1.2 Earlier Works

- Clothiaux et al. [2] proposed identifying spectral peaks by selecting the longest chain and the maximum sum of power spectra values as the principle
- Anandan et al. [3] proposed a method called adaptive moment estimation (AME) based on SNR and wind shear.
- Sinha et al. [4], [5] proposed the multiparameter cost function (MPCF) method for estimating the wind profile.
- Li et al. [6] proposed an improved moment estimation (IME) method for wind estimation.
- Bhattachai et al. [7] proposed the Viterbi data association (VDA) method to track the wind profile.

In the field of turbulence detection, the traditional denoising method is coherent accumulation (CA)

- **Took and Mandic [8]** proposed the quaternion least mean square (QLMS) for wind vector evaluation based on CLMS.

5

Wind Profile Estimation for K-Band Doppler Radar Based on Mutual Convolution Cost Function Method

Roopesh O R (20323085)
2026-09-21

1. Introduction

2

2. Experimental Setup

6

1. Introduction ▶

1.2 Earlier Works

- **Pei and Ding [9]** proposed the fractional Fourier transform (FRFT) method to make the energy of unstable signals more concentrated to improve the filtering performance of adaptive filtering for unstable signals.
- **Chen et al. [10]** proposed the normalized leakage LMS (NL-LMS) to maximize the energy of the target signal and suppress clutter in the FRFT domain.
- **Shi et al. [11]** proposed shrinkage widely linear CLMS (SWL-CLMS), which significantly improved the convergence speed and reduced the steady-state misalignment.
- **Menguc and Acir [12]** proposed the augmented complex valued least mean kurtosis (ACLMK) filter, which utilizes augmented statistics to define the negative kurtosis of the complex error signal as a cost function.
- **Zhang et al. [13]** proposed a wide linear complex estimation input adaptive filter (WLC-EIAF), designed to achieve unbiased estimation in scenarios where input and output signals are affected by noise interference.

2. Experimental Setup

- Experiment equipment: K-band Vortex Radar (VORTRAD)
- Ground truth: LIDAR (usually has 99% accuracy in clear sky)
- At the end, correlation among LiDAR and Radar is obtained



Figure 2: Experiment Setup

7

1. Introduction ▶

1.3 About Proposed Method

- Paper proposes a new method to estimate wind profiles from doppler radars by combining several methods including MCCF. [1]
- Works well under low SNR and multiple peaks conditions.
- The paper primarily focuses K-Band radar and near surface wind profile and on its potential application to efficient wind energy generation.

3.1 Signal Preprocessing

3.1.1 Coherent Accumulation

The CA adds and averages short segments of radar signals improving SNR. Result after N-order CA is represented as:

$$X = [x_{c1}^T, x_{c2}^T, \dots, x_{c(N-1)}^T, x_{cN}^T]$$

Where x_{ci} is result of i^{th} CA.

3.1.2 Slow Sampling

- On every Pulse Repetition Interval (PRI), a pulse is transmitted, radar receives echoes from all range bins (heights)
- For a single height, there will be 1 echo sample from every transmitted pulse.
- To obtain this, a row vector from X is used. This process is called *slow sampling* in the distance dimension

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3.1 Signal Preprocessing

3.1.3 DC and Clutter Removal

- VORTRAD produces large DC power and ground clutter due to its zero-IF RF transceiver architecture.
- Here these are removed.
- The paper doesn't mention a particular clutter removal method

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2.1 Overview

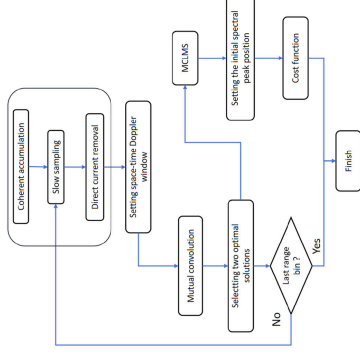


Figure 3: Flow diagram of method given by authors

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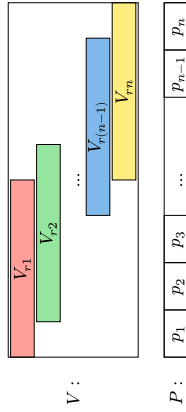
3. Proposed Method

3.3 Mutual Convolution

- Main idea is to use the temporal correlation of echoes to suppress uncorrelated noise.

Method:

1. Take a row vector $P = [p_1, p_2, \dots, p_N]$ of one range bin from X
2. Create new vectors from P with equal length (say half size) $V_{r1} = [p_1, p_2, \dots, p_{N/2}]$
3. From this a new vector V can be written as: $V = [V_{r1}, V_{r2}, \dots, V_{rm}]$ Where V_{ri} is segment of datas from different intervals in same range bin



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3.3 Mutual Convolution

4. Mutual convolution is performed:

$$\begin{aligned} V_{C1} &= V_{r2} * V_{r1}^{\text{HR}} \\ V_{C2} &= V_{r3} * V_{r1}^{\text{HR}} \\ &\vdots \end{aligned}$$

$$V_{C(n-1)} = V_{rm} * V_{r1}^{\text{HR}} \quad x^{\text{HR}} \rightarrow \text{take conjugate and then reverse order}$$

5. Extracting the middle of the vector (To "maintain the same time complexity and velocity resolution of calculation"):

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3.2 Setting Space-Time Doppler Window

- Idea is to restrict search region in both space and time (ST-W) for finding candidate peaks
- Assumes wind velocity changes smoothly with altitude (space) and between successive radar scans (time).
- In the figure
 - ▶ Red $\rightarrow$ Space window
 - ▶ Black $\rightarrow$ Time window

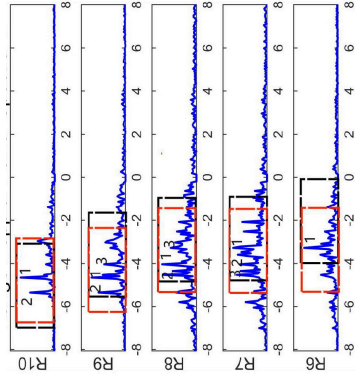


Figure 4: ST-W Window

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3.2 Setting Space-Time Doppler Window

$$\text{Space window: } W_1 = (F_0 - \alpha d_f, F_0 + \alpha d_f)$$

$$\text{Time window: } W_2 = (T_0 - \alpha d_f, T_0 + \alpha d_f)$$

$$\text{Space Time window: ST-W} = \begin{cases} W_1 \cap W_2 & \text{if } W_1 \cap W_2 \geq \alpha d_f \\ W_1 & \text{else} \end{cases}$$

Where

$$\text{wind shear } d_f = \frac{V_d}{d_v} = \frac{V_d}{f_{\text{dmin}} \lambda / L}$$

 V_d = maximum value of wind shear F_0 and T_0 : Space and time spectrum estimation result of the previous range bin L = number of points in FFT. α = window coefficient

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3.4 MCLMS

$$X = \begin{cases} \text{zeros}(1, k-j), & X_r(1:j), & j < k \\ X_r(j-k+1:j), & & j \geq k \end{cases} \quad j \in \{1, 2, \dots, N/2\}$$

(k - length of "coefficient vector")

3. Initialize filter coefficients as $W_1 = 0$. The coefficients are then updated using:

$$Y_j = W_1 X^T$$

$$e_j = X'_{rj} - Y_j$$

$$W'_1 = 0.98W_1 + u e_j X^H$$

4. Repeat above steps until all data are filtered, the coefficients have been fully updated

5. We will use this coefficient to obtain the filtering result Y_r

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3.3 Mutual Convolution

$$[V'_C] = \begin{bmatrix} V'_{C1} \\ V'_{C2} \\ \vdots \\ V'_{C(n-1)} \end{bmatrix} = \begin{bmatrix} V_{C1} \left(\frac{N}{4} : \frac{3N}{4} \right) \\ V_{C2} \left(\frac{N}{4} : \frac{3N}{4} \right) \\ \vdots \\ V_{C(n-1)} \left(\frac{N}{4} : \frac{3N}{4} \right) \end{bmatrix}$$

6. Then maximum of power spectra of each spectrum is computed:

$$V'_{cf} = \left[\max \left\{ \left| \mathcal{F}(V'_C) \right| \right\} \right]$$

7. We take a optimal solution X_r from convolution results based on maximum power V'_{cf} in . Also convolution results of adjacent delays are recorded as suboptimal solutions X'_r

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3.5 Generating the Cost Function

- To find out *candidate peaks*.

Steps

1. Calculate the average value corresponding to the power spectrum of the first range bin and pick out the prominent peak:

$$M_{1,1} = \max \left| \frac{1}{n-1} \sum_{i=1}^{n-1} \mathcal{F}(V'_{Ci}) \right|$$

Use this as starting point for ST-W

2. Compute doppler spectrum of each height bin

$$M_r = |\mathcal{F}(Y_r)|$$

3. In each ST-W, spectral peaks are sorted and up to 3 peaks are selected.

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3.4 MCLMS

- An "adaptive filter", pretrained.
- X_r and X'_r are taken input and desired signal of the filter.
- After the coefficients have been fully updated, filter can recognize stable signals and removes multiple strong clutter.

For finding Filter coefficients:

1. Calculate the normalized parameters of filter coefficients as:

$$u = 1 / \left(\sum_{j=1}^{N/2} |X_r(j)|^2 \right)$$

2. Construct " $\hat{X}$ " from the samples of X_r :

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4.1 Performance at Different Delay Times

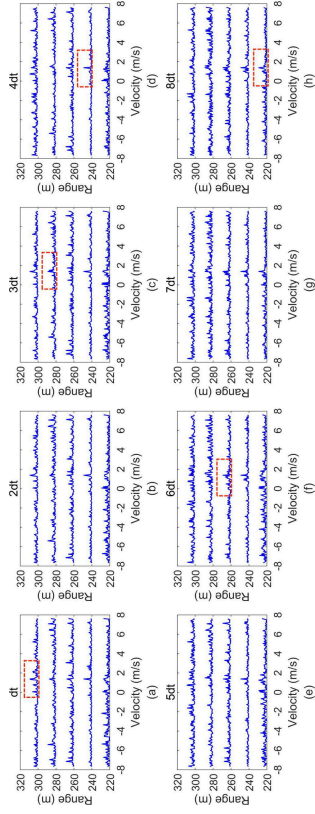


Fig: Optimal solutions are marked by red dotted box of different range bin after mutual convolution (a)–(h)
Represent the filtering effect of dt – 8dt delay time respectively.

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3.5 Generating the Cost Function

4. If $M_{r_{\max}}$ is spectral peak, then candidate peaks are:

$$M_{r,j} \geq 0.4M_{r_{\max}}$$

5. Cost value for each peak is:

$$V_{r+1,j} = 1 - \left| \frac{(M_{r+1} - M_r)d_v}{\alpha V_d} \right| \quad [d_v = \text{velocity resolution}]$$

6. MCCF cost function:

$$F_{r+1,j} = w_1 V_{r+1,j} + w_2 M_{r+1,j} \quad \text{and } w_1 + w_2 = 1$$

w_1, w_2 are constraint weights

7. The peak with the highest cost value is selected

8. Repeat this for all height bins

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4.1 Performance at Different Delay Times

- As the delay time increases, the number of statistics gradually decreases
- Still not possible to determine a specific optimal delay time.

Delay	R1	R2	R3	R4	R5	R6	R7	R8
d_t	1543	1503	1568	1471	1486	1488	1498	1568
$2d_t$	444	404	430	420	398	387	406	403
$3d_t$	179	203	195	224	214	218	223	196
$4d_t$	128	129	91	119	114	127	122	126
$5d_t$	70	72	68	73	85	80	78	70
$6d_t$	39	63	58	69	75	71	66	35
$7d_t$	42	60	38	63	59	51	48	47
$8d_t$	55	66	52	61	69	68	59	55

Table 1: Statistics of optimal delay time for mutual convolution

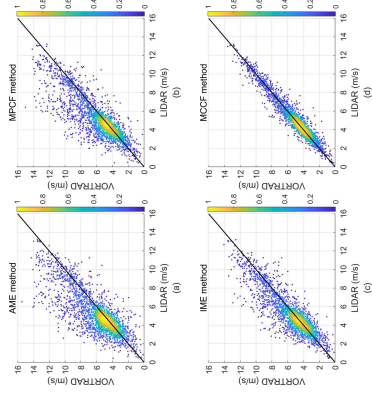
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4. Results

4.4 Correlation with LiDAR observation

- IME shows some improvements over the AME and MPCF
- However MCCF aligns noticeably well with the LiDAR
- Correlation coefficients are summarized below:

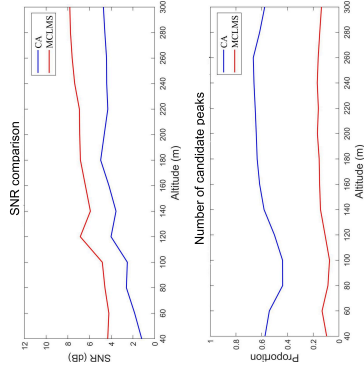
Height(m)	AME	MPCF	IME	MCCF
40	0.8417	0.8368	0.8369	0.9576
80	0.8319	0.8350	0.8461	0.9609
120	0.8320	0.8431	0.8695	0.9625
160	0.8137	0.8353	0.8761	0.9641
200	0.7908	0.8276	0.8764	0.9647
240	0.7830	0.8202	0.8738	0.9650
280	0.7837	0.8219	0.8744	0.9651



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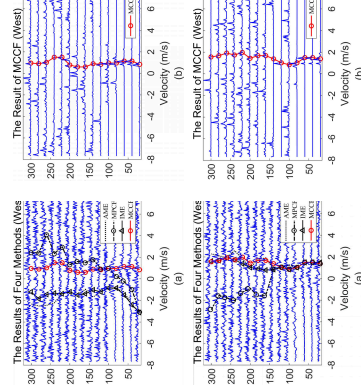
4.2 Comparison of Peak Number and SNR

- Average SNR for all bins plotted for CA and proposed MCLMS filters at different altitudes
- Proportion of spectra containing more than six candidate peaks



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4.3 Wind Profile Estimation Results



- Comparison of four methods based on the data recorded on
- Dec 18, 2022 (15:19:32) (top row)
 - Dec 18, 2022 (15:19:33) (bottom row)

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5. Conclusion & Limitations

6. References

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5. Conclusion & Limitations

- The proposed algorithm was applied on data collected by the VORTRAD K-band wind profiler and verified against LiDAR observation
- Method reduced false spectral peaks and improved the SNR ratio by approximately 2–4 dB
- Algorithm produces smoother and more continuous wind profiles
- Achieves highest correlation coefficient (0.96) with LiDAR observation.
- Authors note that the current method is validated only for clear-weather conditions and does not account for precipitation echoes such as rain or snow

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Thank You